

Tokenizer and Classifications

- **Text Classification:** Assigning categories to text, such as sentiment labels (positive/negative) or grammatical correctness (acceptable/unacceptable).
- **Challenges in Text Classification:** Addressing ambiguity, sarcasm, irony, and multilingual texts, which can complicate classification tasks.

Using Hugging Face's Pipeline: You practiced using the `pipeline` function from the Hugging Face `transformers` library for text classification tasks. For example, to check grammatical correctness, you used:

```
classifier = pipeline(task="text-classification",  
model="abdulmatinomotoso/English_Grammar_Checker")  
output = classifier("I will walk dog")
```

-
- **Zero-Shot Classification:** Learning about the capability of models to classify text into categories they weren't explicitly trained for, using a list of candidate labels.

Work with LLMOps

- **Governance:** Involves policies, guidelines, and frameworks that govern the development, deployment, and usage of LLM applications.
- **Security:** Measures to prevent unauthorized access, data breaches, adversarial attacks, and misuse of the models' outputs.
- **Role-Based Access Control (RBAC):** Assigns permissions to roles, and users are assigned to those roles, ensuring appropriate access.
- **Zero Trust Security Model:** Requires all users to be authenticated, authorized, and continuously validated.
- **Common Threats:**
 - **Prompt Injection:** Attackers manipulate input fields to execute unauthorized commands.
 - **Output Manipulation:** Alters an LLM's output, potentially leading to downstream attacks.
 - **Denial-of-Service (DoS):** Users flood the application with requests, causing performance issues.
 - **Data Poisoning:** Injecting false or harmful data into the training set.
- **Mitigation Strategies:**
 - Avoid revealing sensitive information in prompts.
 - Use trusted data sources and implement filters.
 - Limit request rates and manage resource usage.

